A new actual twist transfer over $\mathbf{F}_{37}$

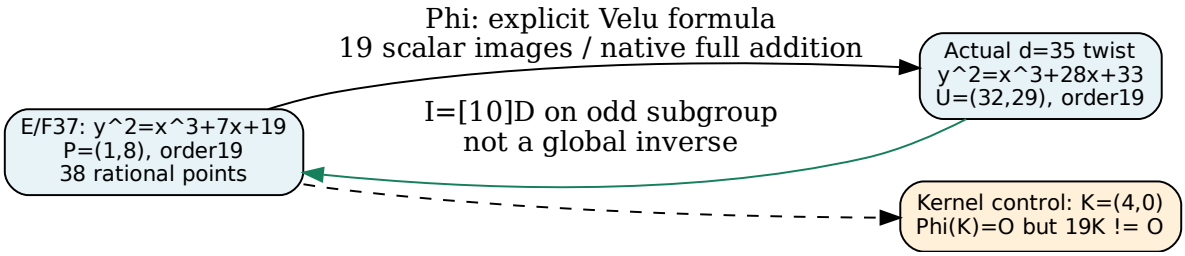
2026-10-30 PR2381 fixed public synthetic fixture

$E : y^2 = x^3 + 7x + 19$, $T : y^2 = x^3 + 28x + 33$, the actual nonsquared $= 35$twist. Each has38rationalpoints. The fixed lexicographic selection gives $P = (1, 8)$, $U = (32, 29)$ of order19. No registered curve alias or end-ring classification is inferred.

$$\Phi(x, y) = \left(x + \frac{18}{x - 4},\ y\left(1 - \frac{18}{(x - 4)^2}\right)\right),$$

$$D(x, y) = \left(28\left(x + \frac{35}{x - 29}\right),\ 14y\left(1 - \frac{35}{(x - 29)^2}\right)\right).$$

Origins and the respective kernels$(4, 0)$,$(29, 0)$maptoorigin. The actual inverse on the19subgroup is $I = [10]D$.



Independent native verification: all38+38pointmaps,both compositions= [2], 2888additionpairs and19scalarimage Wrong[9]Dfails on18nonzero points.

Kernel result:19publicnamed/22audited declarations,zero failing/extra trusted/native axioms. Actual Mathlib nonsingular points and coordinate formulas, not logtables. Lean checks equations/order/scalartransport/subgr inverses and necessary coefficient obstruction to a short-model change. For arbitrary actual source$Q$with explicit cyclic membership and$n \in \mathrm{Fin}\, 19$,

$$(\exists i \in \mathrm{Fin}\, 19,\ Q = iP) \quad \Longrightarrow \quad [\Phi(Q) = nU \iff Q = nP].$$

This doesnot reveal the unknown index to a solver or assert generic geometric classification. Globalinverse is false at the source2torsion point.

Full source-policy frontier,not a favorable prefix

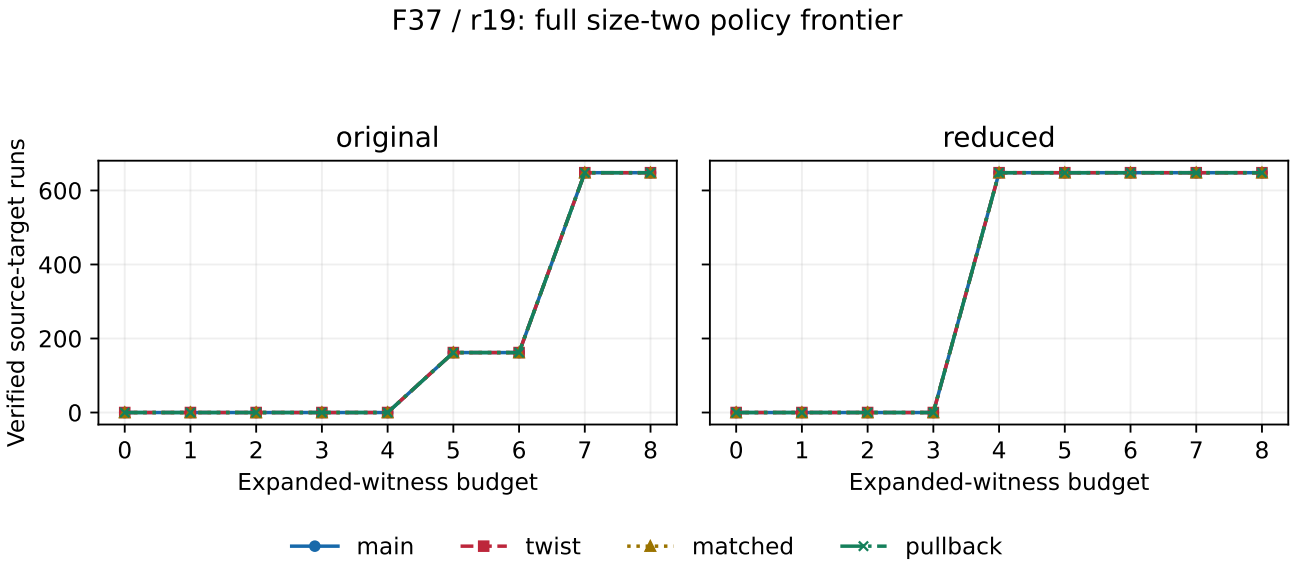


Figure. Native-emitted counts only.36policies×18sourcepoints perrole/cap; curves may overlap. Exact finite values,no timing or sampling uncertainty inferred. All144arms (including matched/pullback),twoencodings,caps0–8:2592configurations, 46656separate attempts repeating18sourcepoints,not46656distinctchallenges.

Exact independently reconstructed outcome	Count
Verified original-source recoveries	19440
Blocked public-log setups	1512
Policy/target/budget comparison records	3779136
Higher rank/lower requested cap,recovery admitted	55230
Beating strongest canonical source policy	0
Changed controls rejected101/no receipt	10
Backend tests passed	22

55230counts pairrecords,not unique targets or wins. Lower requested budget neednot imply lower actual solving degree. None survives the strongest-source comparison. No practical speedup,stationary usefulyield,intrinsicregularity or cross-field performance gain is established.

What was run,what remains **Actual algebraic instrument.** NewF37/r19version,not the hard-codedF13binary. Three inputs:$S_3$and both product-membership constraints. Flat Macaulayrows carry exact original-input witnesses. A unit or bounded Buchbergercertificate must pass BEFOREextraction. Above-bound shifts are denied BEFOREconstruction; capunknown withholdsroots/rows. Retain a smalleralreadyfoundwitness only for an IDENTICALpolynomial,localtoonesolve. Matrix work still runs;no cost claim.

Publicsetup derives logs ONLY from admittedrelations and checksactualbasepoints; incomplete rankblocksalltargets. Recovery uses actual unknown$Q$,freshshifted queries,allfailed/identityshifts/signlifts,and verifies the scalar on ORIGINALE/P. Outputreferencebinding to publicqueryrecords happensafterfreshsolve and isunpriced; it never supplies theanswer. Directenumeration isseparategenericdecomposition, not an algebraic degree reference. Setup,allocation,group/field and verification work are not a complete calibrated price.

Preserved evidence. Complete C++/Rust transfer and2592case replays pass. Falseinverse plus ninecase corruptions genuinely change/re-hash and reject101without receipts;22backendtests pass. Canonical cases/indexes,ALL tree reconstruction,raw source/argv/exits/logs/proofreceipts and canonicalexport remain. All old18IDC1records/proof are untouched. No new universal polynomial identity or fresh combined/global proof audit is claimed.

FirstLean setup failed cloning a newMathlibcheckout before elaboration. The same pinned$d13f23\ldots$source was reused from theexistingignoredcache with the gate unmodified. Actual gateexit0/statusPASS/22audit was reconfirmed after an outerPTY exit discrepancy;no hypothesis/data changed. FrozenJSON's copied F13ordering phrase has a datedmetadataerratum;newF37scope was already explicit in question, arms,Markdown,source and coverageassertions. No posthoc selection or statistical confirmation claim.

Next: complete sizeTHREEbases over this verifiedroute (84canonicalpolicies percurve plus controls),after a new fixed protocol. Cubicmembership leavesS_3 degree$\leq$ 2pervariable unchanged:declare encodingequality ratherthan a falsecontrast. That study is not yet implemented/executed. Original advantage remains OPEN.

KeepPRopen/non-draft,auto-mergeOFF.1921registrations,root5140suite600s/124incomplete, naming,requiredCI/revi and broadergeometry/ probability/regularity/fullcost remainheld. Existinglocal2workers,600seccommands, window13:40–14:40UTC,no paidresources/arbitraryuploads. Source/evidence/hash/ primarySage/Lean/canonicalexp links are in adjacentREADME.